

Record Gain-Bandwidth Product Si-Ge Avalanche Photodiode Enabling an 8×300 Gbps DWDM Receiver

Chao Cheng^(1,2), Xishan Yu^(1,2), Qian Liu^(1,2), Xianglin Bu^(1,2), Shenlei Bao^(1,2), Jintao Xue^(1,2), Wenfu Zhang^(1,2), Binhao Wang^(1,2)

⁽¹⁾ The State Key Laboratory of Transient Optics and Photonics, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an 710119, China, wangbinhao@opt.ac.cn

⁽²⁾ University of Chinese Academy of Sciences, Beijing 101408, China.

Abstract

We demonstrate a Si-Ge avalanche photodiode with a record 9460 GHz gain-bandwidth product via inductive peaking, achieving a 55 GHz bandwidth and a peak gain of 172. Integrated with dual-ring filters on an SOI platform, it enables an 8×300 Gbps PAM4 DWDM receiver.

Introduction

Driven by emerging artificial intelligence (AI) applications, data centers and high-performance computing (HPC) systems are experiencing exponential growth in communication capacity and bandwidth demands. Conventional electrical interconnects are approaching fundamental limits in bandwidth density and energy efficiency, making them insufficient to meet future computing requirements [1, 2]. Silicon photonics (SiPh), offering compatibility with complementary metal-oxide-semiconductor (CMOS) fabrication and high bandwidth density, has emerged as a promising solution. In particular, co-packaged optics (CPO) and optical input/output (OIO) architectures are being actively explored for next-generation optical interconnects [3, 4].

Within SiPh receivers, silicon-germanium (Si-

Ge) avalanche photodiodes (APDs) are preferred over conventional photodiodes due to their internal gain from impact ionization, which relaxes transimpedance amplifier (TIA) requirements and improves link budgets. The gain-bandwidth product (GBP), governed by the ionization coefficient ratio k , is the central figure of merit for high-speed APDs: a higher GBP simultaneously permits greater gain at a given bandwidth or wider bandwidth at a given gain. Although the GBP of Si-Ge APDs has progressed rapidly in recent years, from 340 GHz to 7564 GHz [5, 6], achieving GBPs beyond this level while maintaining compatibility with standard silicon-on-insulator (SOI) fabrication and dense wavelength-division multiplexing (DWDM) integration remains an open challenge. In particular, junction capacitance imposes a hard bandwidth ceiling that has not been sys-

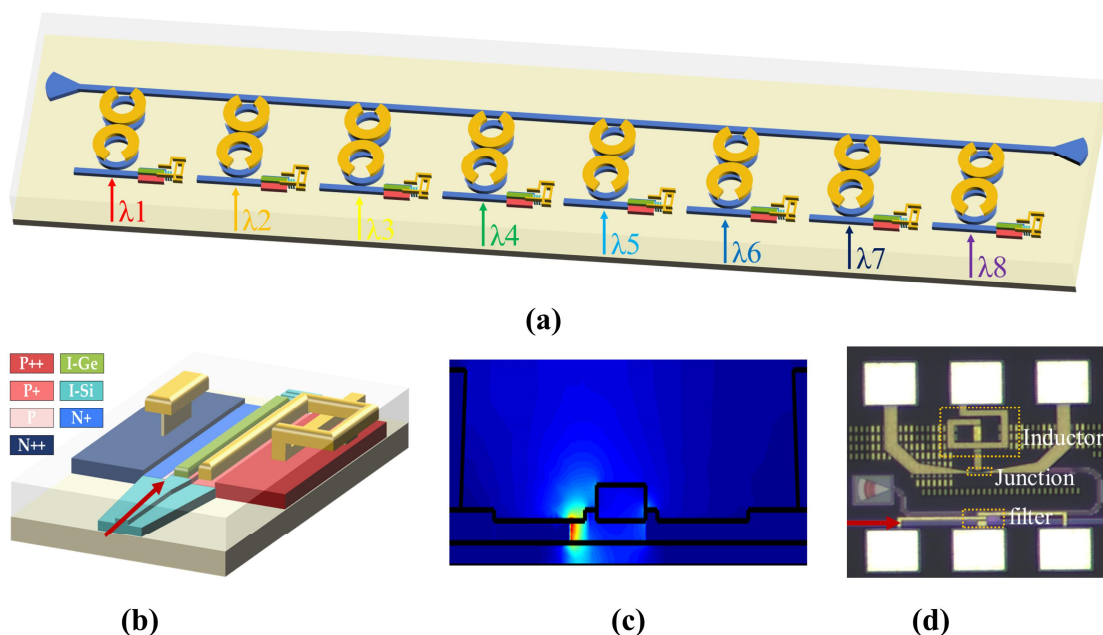
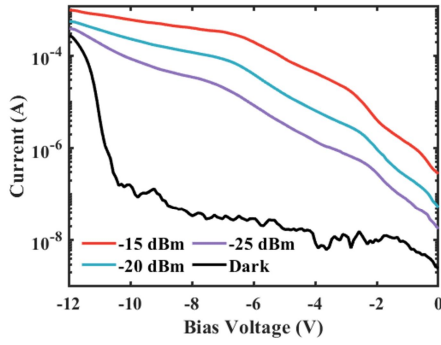
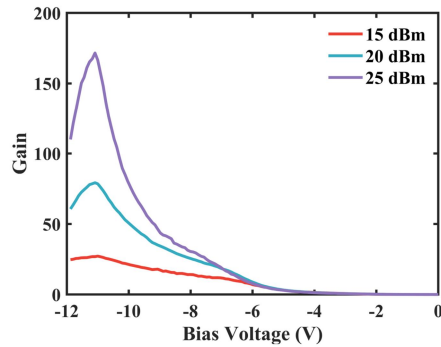


Fig. 1: (a) Schematic of the proposed 8-channel DWDM receiver; (b) 3D schematic of the Si-Ge APD; (c) Cross-sectional electric field profile of the Si-Ge APD at the operating voltage; (d) Microscopic image of the single-channel receiver.



(a)



(b)

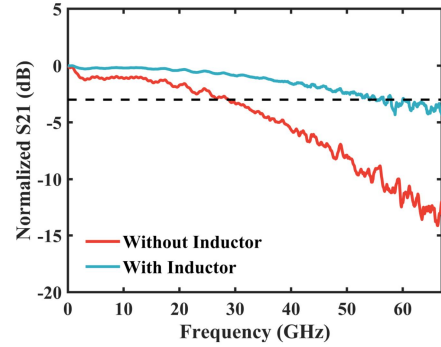
Fig. 2: (a) Measured I-V curves under different input optical powers; (b) APD gain as a function of bias voltage and input optical power.

tem atically addressed in prior DWDM receiver demonstrations.

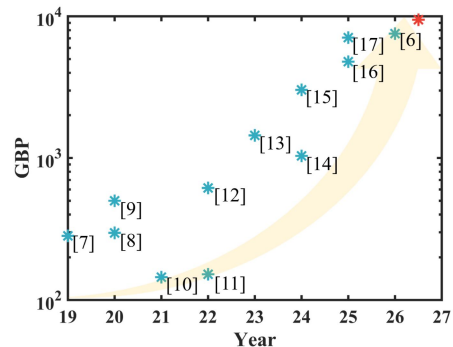
In this work, a separate-absorption-charge-multiplication (SACM) structure is employed to spatially separate the absorption and avalanche regions, thereby reducing dark current and enhancing gain. Meanwhile, on-chip inductive peaking is introduced to compensate for bandwidth limitations imposed by junction capacitance. As a result, a Si-Ge APD with a GBP of up to 9460 GHz is achieved, featuring a 3 dB bandwidth of 55 GHz and a peak gain of 172. We further integrate this APD with 200 GHz-spaced dual-ring filters and grating couplers (GCs) on a standard SOI platform to realise an 8-channel DWDM receiver. The receiver demonstrates 300 Gbps PAM4 operation with clear eye openings across all eight channels, achieving an aggregate data rate of 2.4 Tbps per fiber in a compact footprint of $2.9 \text{ mm} \times 0.38 \text{ mm}$. To the best of our knowledge, this represents the highest GBP reported for a Si-Ge APD integrated within a DWDM receiver. The proposed architecture provides a scalable and compact solution for high-sensitivity, high-capacity optical interconnects.

Device Structure

Fig. 1(a) presents a schematic of the proposed receiver, integrating GCs, dual-ring filters, and Si-



(a)



(b)

Fig. 3: (a) APD 3 dB OE bandwidth with and without inductive peaking; (b) Evolution of the GBP of Si-Ge APDs.

Ge APDs. Optical signals are coupled into the Si waveguide via the GCs and demultiplexed by the dual-ring filters with 200 GHz channel spacing. Fig. 1(b) presents a 3D schematic of the APD. The device employs a lateral p^+i-p-n^+ SACM structure, in which O-band optical signals are evanescently coupled into the Ge absorption region for photodetection. To overcome bandwidth limitations imposed by junction capacitance, a 200 pH inductor is integrated to implement inductive peaking, compensating parasitic effects and extending the high-frequency response. Fig. 1(c) illustrates the cross-sectional electric field distribution of the Si-Ge APD at the operating voltage. Through optimized doping design, the high electric field is effectively confined within the Si region, leveraging Si's k to suppress excess noise. Fig. 1(d) shows a microscopic image of the single-channel receiver. The dual-ring filter exhibits a 1 dB optical bandwidth of 90 GHz, providing sufficient tolerance for high-speed modulation formats. Its estimated insertion loss of ~ 0.5 dB per channel is effectively compensated by the APD's high gain, maintaining an adequate signal-to-noise ratio.

As a result, both the bandwidth and gain are improved, leading to a significantly enhanced GBP. This design provides a viable solution for high-speed and high-sensitivity detection under low optical power conditions.

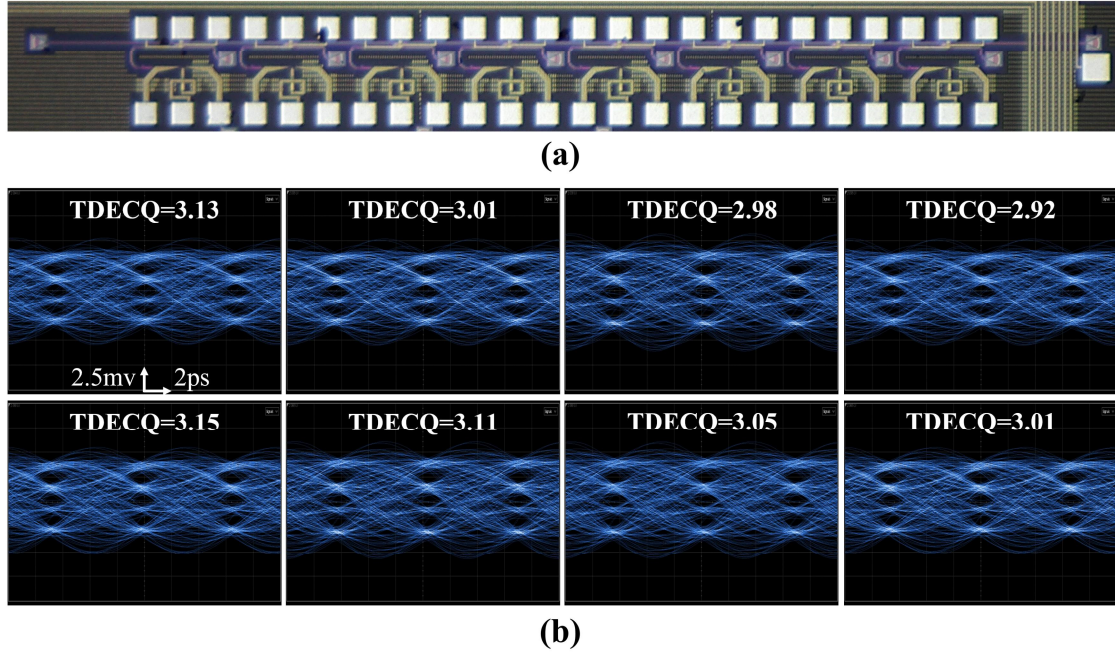


Fig. 4: (a) Microscopic image of the eight-channel DWDM receiver; (b) 300 Gbps PAM4 eye diagrams of all eight channels.

Experimental Results

Fig. 2(a) shows the I-V characteristics of the APD under varying reverse bias voltages, including both dark conditions and incident optical powers ranging from -25 dBm to -15 dBm. At a reverse bias of 11 V, the dark current remains as low as $3.89 \mu\text{A}$. Fig. 2(b) presents the APD gain as a function of reverse bias for different incident optical powers. The gain initially increases with reverse bias, peaking at values of 27, 79 and 172 at 11 V for incident powers of -15 , -20 and -25 dBm, respectively, and then declines as the breakdown voltage is approached due to the rapid rise in dark current.

Fig. 3(a) shows the measured 3 dB optoelectrical (OE) bandwidth, comparing the performance of the APD with inductive peaking at the operating voltage to that of a previously reported APD without inductive peaking [6]. The results indicate that, at a reverse bias voltage of 11 V, the APD achieves a 3 dB OE bandwidth of 55 GHz, representing approximately a 90% improvement over an APD without inductive peaking. Fig. 3(b) summarizes reported GBPs of Si-Ge APDs in recent years, illustrating that higher GBPs remain a primary research focus [6–17]. The APD demonstrated in this work achieves a GBP of 9460 GHz (gain $\times$ bandwidth = 172×55 GHz) at 11 V, highlighted in red in Fig. 3(b), representing one of the highest values reported to date.

We further performed eye diagram measurements on the eight-channel receiver, with its microscopic image shown in Fig. 4(a). The experimental setup employed a tunable laser providing continuous-wave (CW) light, which was first adjusted by a polarization controller (PC) and

then modulated by a 67 GHz commercial thin-film lithium niobate (TFLN) modulator. The modulation signal was generated by a Keysight arbitrary waveform generator (M8199B) with a sampling rate of 256 GSa/s. The modulated signal was subsequently attenuated by a variable optical attenuator (VOA), passed through a PC, and finally coupled into the APD for photodetection. Fig. 4(b) presents the eye diagrams of all eight channels at a data rate of 300 Gbps PAM4. The transmitter dispersion eye closure quaternary (TDECQ) values range from 2.92 to 3.15 dB, indicating excellent signal integrity and channel uniformity.

Conclusions

In this work, a Si-Ge APD with a GBP of 9460 GHz was realized through inductive peaking, setting a new performance record. The device exhibits a 3 dB OE bandwidth of 55 GHz and a peak gain of 172 at 11 V reverse bias, representing approximately 90% bandwidth enhancement compared to the same structure without inductive peaking. By integrating these high-performance APDs with GCs and dual-ring filters on a standard SOI platform, we developed an 8×300 Gbps SiPh DWDM receiver. The receiver successfully produced clear open eye diagrams for all eight channels, occupying an ultra-compact footprint of only $2.9 \text{ mm} \times 0.38 \text{ mm}$, demonstrating single-fiber transmission at an aggregate rate of 2.4 Tbps. The demonstrated combination of record GBP, wide bandwidth, and DWDM system compatibility establishes a strong foundation for next-generation high-capacity optical interconnects in AI and HPC environments.

References

- [1] J. Xue et al., "A 4×112 gb/s ultra-compact polarization-insensitive silicon photonics wdm receiver with cmos tia for co-packaged optics and optical i/o", *Journal of Lightwave Technology*, vol. 42, no. 17, pp. 6028–6035, 2024. DOI: 10.1109/JLT.2024.3402053.
- [2] B. Wang and J. Mu, "High-speed si-ge avalanche photodiodes", *PhotonIX*, vol. 3, no. 1, p. 8, 2022. DOI: 10.1186/s43074-022-00052-6.
- [3] B. Buscaino, E. Chen, J. W. Stewart, T. Pham, and J. M. Kahn, "External vs. integrated light sources for intra-data center co-packaged optical interfaces", *Journal of Lightwave Technology*, vol. 39, no. 7, pp. 1984–1996, 2021. DOI: 10.1109/JLT.2020.3043653.
- [4] K. Hosseini et al., "8 tbps co-packaged fpga and silicon photonics optical io", in *2021 Optical Fiber Communications Conference and Exhibition (OFC)*, 2021, pp. 1–3.
- [5] Y. Kang et al., "Monolithic germanium/silicon avalanche photodiodes with 340 ghz gain–bandwidth product", *Nature photonics*, vol. 3, no. 1, pp. 59–63, 2009. DOI: 10.1038/nphoton.2008.247.
- [6] J. Xue, C. Cheng, S. Bao, Q. Liu, W. Zhang, and B. Wang, "High-speed si-ge avalanche photodiode with a gain-bandwidth product of 7564 ghz", *Nature Communications*, 2026. DOI: 10.1038/s41467-026-70461-9.
- [7] X. Zeng, Z. Huang, B. Wang, D. Liang, M. Fiorentino, and R. G. Beausoleil, "Silicon–germanium avalanche photodiodes with direct control of electric field in charge multiplication region", *Optica*, vol. 6, no. 6, pp. 772–777, 2019.
- [8] S. A. Srinivasan, M. Berciano, P. De Heyn, S. Lardinois, M. Pantouvaki, and J. Van Campenhout, "27 ghz silicon-contacted waveguide-coupled ge/si avalanche photodiode", *Journal of Lightwave Technology*, vol. 38, no. 11, pp. 3044–3050, 2020. DOI: 10.1109/JLT.2020.2986923.
- [9] B. Wang et al., "64 gb/s low-voltage waveguide sige avalanche photodiodes with distributed bragg reflectors", *Photonics Research*, vol. 8, no. 7, pp. 1118–1123, 2020. DOI: 10.1364/PRJ.390339.
- [10] X. Hu et al., "High-performance germanium avalanche photodetector for 100 gbit/s photonics receivers", *Optics Letters*, vol. 46, no. 16, pp. 3837–3840, 2021. DOI: 10.1364/OL.433694.
- [11] T. Okimoto et al., "106-gb/s waveguide alinas/gainas avalanche photodiode with butt-joint coupling structure", in *2022 Optical Fiber Communications Conference and Exhibition (OFC)*, 2022, pp. 01–03.
- [12] Y. Xiang, H. Cao, C. Liu, J. Guo, and D. Dai, "High-speed waveguide ge/si avalanche photodiode with a gain-bandwidth product of 615 ghz", *Optica*, vol. 9, no. 7, pp. 762–769, 2022. DOI: 10.1364/OPTICA.462609.
- [13] H. Cao, Y. Xiang, W. Sun, J. Xie, J. Guo, and D. Dai, "Waveguide ge/si avalanche photodetector with ultra-high gain-bandwidth product of 1440ghz", in *2023 Asia Communications and Photonics Conference/2023 International Photonics and Optoelectronics Meetings (ACP/POEM)*, 2023, pp. 1–3. DOI: 10.1109/ACP/POEM59049.2023.10368970.
- [14] Y. Shi et al., "Avalanche photodiode with ultrahigh gain–bandwidth product of 1,033 ghz", *Nature Photonics*, vol. 18, no. 6, pp. 610–616, 2024. DOI: 10.1038/s41566-024-01421-2.
- [15] H. Cao, W. Sun, Y. Xiang, J. Xie, J. Guo, and D. Dai, "Ultrahigh gain-bandwidth product ge/si avalanche photodetector assisted by nonuniform electric field distribution", in *ECOC 2024; 50th European Conference on Optical Communication*, 2024, pp. 1964–1967.
- [16] C. Cheng, J. Xue, S. Bao, Q. Liu, W. Zhang, and B. Wang, "A 112 gbps silicon-germanium avalanche photodiode with ultra-high gain-bandwidth product", in *2025 Optical Fiber Communications Conference and Exhibition (OFC)*, 2025, pp. 1–3.
- [17] Y. Shi, M. Zou, Z. Li, Y. Yu, and X. Zhang, "Ultrafast avalanche photodiode exceeding 100 ghz bandwidth", *Nature Communications*, vol. 16, no. 1, p. 11 058, 2025. DOI: 10.1038/s41467-025-66047-6.